



Accenture Network Security — Tier 3

Comprehensive Theory Guide & Practice Questions with Integrated Solutions

We'll cover:

- Zero Trust
- Honeypot
- Security Monitoring
- SIEM
- Antivirus / Anti-malware
- Proxy Firewall
- Next-Generation Firewall (NGFW)
- Security Auditing & Logging
- Vulnerability vs Threat vs Risk
- Patch Management
- Principle of Least Privilege
- Defense in Depth
- Security Incident & Data Breach
- 25 Accenture Practice MCQs with Inline Solutions

1. Zero Trust ★★★★★

Definition

Zero Trust is a security model based on the principle of "**never trust, always verify.**"

Traditional approach:

```
Internet
  ↓
Firewall
  ↓
Trusted Internal Network
  ↓
Users
```

The assumption was roughly: *Internal users/devices are trusted.*

Zero Trust changes this:

```
User
  ↓
Verify identity
  ↓
Verify device
  ↓
Check permissions
  ↓
Allow limited access
  ↓
Continuously monitor
```

Core idea

Being inside the network does not automatically make someone trustworthy.

Example

An employee is sitting inside the company's office.

Traditional thinking: They're inside the office → probably trusted.

Zero Trust: Being inside the office isn't enough. Verify the user and device before granting access.

Important Zero Trust principles

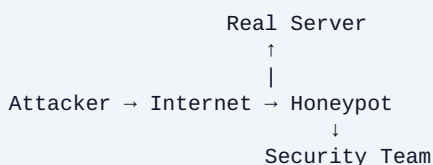
1. **Verify explicitly:** Verify things such as User identity, Device, Location, Application, and Risk level.
2. **Least privilege:** Give users only the permissions they need.
Example: Employee needs payroll report → Give access to payroll report. Don't give full database access unless necessary.
3. **Assume breach:** Design systems with the assumption that an attacker may already be inside. Therefore, continuously monitor and limit access.

 **Keyword:** Never trust, always verify → **Zero Trust**

2. Honeypot ★★★★★

Definition

A honeypot is a deliberately designed decoy system or resource used to attract, detect, and study attackers. *Think: A trap for attackers.*



The attacker thinks: "I found an important server." But actually: *It's a monitored decoy.*

Why use a honeypot?

- Detect attackers
- Study attack techniques
- Collect attack information
- Divert attackers away from real systems

Example: A company creates fake-admin-server.company.com. It looks valuable. An attacker attempts to access it. Security team receives an alert. That's a honeypot scenario.

Honeypot vs Firewall

Honeypot	Firewall
Decoy system	Traffic-control / security boundary
Attracts / detects attackers	Allows / blocks traffic
Used for detection / research	Used for access control

 **MCQ Keyword:** Decoy / trap / fake server → **Honeypot**

3. Security Monitoring ★★★★★

Definition

Security monitoring is the continuous observation and analysis of systems, networks, logs, and events to identify suspicious or malicious activity.

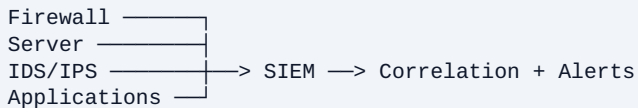
Network Traffic → Logs → Monitoring System → Suspicious activity → Alert → Security Team

What can be monitored? Login attempts, Network traffic, Firewall events, Server logs, Failed authentication, Unusual file access, Malware alerts, and Configuration changes.

Example: A user normally logs in from India. Suddenly: Login from India at 10:00 AM, and Login from another country at 10:02 AM. Security monitoring may flag this as suspicious.

4. SIEM (Security Information and Event Management)

You may encounter SIEM in security MCQs. It collects and analyzes security-related logs/events from different systems.



Why SIEM? Instead of checking Firewall logs, Server logs, and Application logs individually, SIEM brings security events together for centralized analysis.

Accenture Level: SIEM = Collect + analyze security events/logs centrally.

5. Antivirus / Anti-malware ★★★★★

Antivirus: Software designed to detect, prevent, and remove malicious software such as Viruses, Trojans, Some ransomware, Worms, and Spyware.

File → Antivirus → Malicious detected → Block / Quarantine

Quarantine: Isolate suspicious/malicious file so it can't freely execute or affect the system.

Signature-based detection: Compares files against known malware signatures. *Limitation:* May struggle with brand-new zero-day malware without known signatures (modern tools use heuristic/behavior-based detection).

6. Proxy Firewall ★★★★★

What is a proxy? Acts as an intermediary between the client and destination server (Client → Proxy → Internet Server).

Proxy Firewall: Operates at a higher level than basic packet filtering and can inspect application-level information (e.g., allow company.com, block malicious-site.com).

- **Packet filtering firewall:** Looks at packet/network info (IP, Port, Protocol).
- **Proxy firewall:** Acts as an intermediary and inspects application-level communication.

7. NGFW — Next-Generation Firewall ★★★★★

Definition: NGFW is an advanced firewall that combines traditional firewall functionality with additional security capabilities and deeper traffic inspection.

Traditional Firewall: IP, Port, Protocol.

NGFW Additions: Application awareness/control, Intrusion prevention (IPS), Deeper inspection (DPI), URL/web filtering, and Malware-related controls.

Traditional Firewall	NGFW
Basic traffic filtering	Advanced inspection
IP / port / protocol	Application-aware capabilities
Basic access control	Integrated security features (IPS, URL filter)

 **Keyword:** Application-aware + advanced firewall → NGFW

8. Security Auditing & Logging

Logging: Recording system events (e.g., 10:30 — User login, 10:35 — File accessed, 10:40 — Password changed). These records are called **Logs**.

Security Auditing: Examines activities and configurations to determine whether systems comply with security policies and requirements.

Why logs matter: Detecting attacks, investigating incidents, troubleshooting, compliance, and identifying suspicious behavior (e.g., "Who accessed the confidential file at 2:30 AM?").

9. Vulnerability vs Threat vs Risk ★★★★★

- **Vulnerability:** A weakness in a system (e.g., Old software + known security bug). → **Weakness**
- **Threat:** Something that could potentially cause harm (e.g., Hacker, malware, natural disaster). → **Potential danger**
- **Risk:** The potential for loss or damage when a threat can exploit a vulnerability. → **Potential impact/loss**

Threat + Vulnerability → Risk

Example: Server has outdated software (Vulnerability) + Attacker (Threat)
→ Attacker compromises server causing damage (Risk)

10. Patch Management ★★★

Definition: The process of identifying, testing, deploying, and maintaining software/security updates to fix vulnerabilities.

Software v1 → Vulnerability discovered → Vendor releases patch → Admin installs patch → Fixed

11. Principle of Least Privilege ★★★★★

Giving users, applications, and systems only the minimum permissions required to perform their tasks. Minimizes damage if an account is compromised.

 **Keyword:** Minimum required permission → Least privilege

12. Defense in Depth ★★ ★

Using multiple layers of security instead of relying on a single security control. If one layer fails, another still protects the system.

Internet → Firewall → IDS/IPS → Authentication → Access Control → Antivirus → Encrypted Data

13. Security Incident & 14. Data Breach

Security Incident: An event that potentially compromises confidentiality, integrity, or availability (Malware, Unauthorized login, Data breach, DDoS).

Lifecycle: Detect → Analyze → Contain → Eradicate → Recover → Learn/Improve.

Data Breach: Occurs when sensitive or protected information is accessed, disclosed, or exposed without authorization. It is the outcome/exposure, not the attack technique itself.

Accenture Network Security — Tier 3 MCQ Practice (with Inline Solutions)

Q1. Zero Trust

A company does not automatically trust employees simply because they are connected to the internal corporate network. Every request for a sensitive resource must be verified. Which security model is being implemented?

- A. Perimeter Security
- B. Zero Trust
- C. DAC
- D. Open Access

Answer: B. Zero Trust ✓

Explanation: Every access request is verified regardless of whether the user is inside or outside the network boundary.

A. ✗ Relies on trusted network | C. ✗ Discretionary Access Control | D. ✗ Unrestricted access

🧠 **Keyword:** Never trust, always verify.

Q2. Zero Trust Principle

Which principle is most closely associated with Zero Trust?

- A. Trust all internal users
- B. Never trust, always verify
- C. Trust devices inside the firewall
- D. Allow unrestricted internal access

Answer: B. Never trust, always verify ✓

Explanation: Zero Trust removes implicit trust from all users and devices inside or outside the perimeter.

Q3. Honeypot

A security team deploys a fake database containing dummy records. Attackers who attempt to access it are monitored by the security team. What is this system?

- A. Firewall
- B. Honeypot
- C. VPN
- D. Proxy

Answer: B. Honeypot ✓

Explanation: A deliberately designed decoy system deployed to attract and study attackers.

A. ✗ *Filters traffic* | C. ✗ *Secure connection* | D. ✗ *Intermediary*

🧠 **Keyword:** Fake server / decoy = Honeypot.

Q4. Honeypot Scenario

What is the primary purpose of a honeypot?

- A. Increase network bandwidth
- B. Encrypt all network traffic
- C. Attract and detect/study attackers
- D. Assign IP addresses

Answer: C. Attract and detect/study attackers ✓

Explanation: Honeypots act as traps to monitor attacker techniques and divert them from real systems.

Q5. SIEM

A company collects security logs from Firewalls, Servers, IDS/IPS, and Applications and analyzes them centrally to identify suspicious activity. What technology is being used?

- A. SIEM
- B. VPN
- C. AES
- D. DHCP

Answer: A. SIEM ✓

Explanation: SIEM (Security Information and Event Management) centralizes and correlates security events and logs.

🧠 **Keyword:** Centralized log analysis = SIEM.

Q6. Security Monitoring

A security analyst notices that an employee's account has generated 500 failed login attempts within five minutes. Which activity would most likely identify this behavior?

- A. Security monitoring
- B. Data encryption
- C. Compression
- D. NAT

Answer: A. Security monitoring ✓

Explanation: Continuous observation of authentication logs and alerts identifies brute-force attempts.

Q7. Antivirus

An employee downloads a file. Before it executes, security software detects that the file contains known malicious code and blocks it. Which technology is primarily responsible?

- A. Antivirus/anti-malware
- B. VPN
- C. ACL
- D. DNS

Answer: A. Antivirus/anti-malware ✓

Explanation: Antivirus software uses signature matching and heuristic analysis to identify and block malicious code.

Q8. Quarantine

An antivirus application detects a malicious executable and moves it to an isolated location where it cannot execute normally. What is this process called?

- A. Encryption
- B. Quarantine
- C. Authentication
- D. Spoofing

Answer: B. Quarantine ✓

Explanation: Quarantine isolates suspicious files in a safe location so they cannot execute or infect the system.

 **Keyword:** Malware isolated → Quarantine.

Q9. NGFW

A company wants a firewall that can perform traditional traffic filtering while also providing application-aware inspection and additional security capabilities. Which is most appropriate?

- A. Basic hub
- B. NGFW
- C. DHCP server
- D. DNS server

Answer: B. NGFW ✓

Explanation: Next-Generation Firewalls combine packet filtering with Layer 7 application awareness and integrated IPS.

 **Keyword:** Application-aware + advanced firewall → NGFW.

Q10. Proxy Firewall

A security device receives a request from an internal client and forwards the request to the external server on behalf of the client. What is this device most likely acting as?

- A. Proxy
- B. Router only
- C. DNS resolver
- D. DHCP server

Answer: A. Proxy ✓

Explanation: A proxy serves as an intermediary, terminating and re-initiating requests on behalf of the client.

Q11. Vulnerability

A web application contains an unpatched software flaw that attackers can exploit. What does the flaw represent?

- A. Threat
- B. Vulnerability
- C. Risk
- D. Incident response

Answer: B. Vulnerability ✓

Explanation: A software bug, misconfiguration, or flaw in a system is a vulnerability (weakness).

 **Keyword:** Flaw / weakness = Vulnerability.

Q12. Threat

A hacker is actively looking for vulnerable systems to attack. In security terminology, the hacker represents a:

- A. Vulnerability
- B. Threat
- C. Patch
- D. Firewall

Answer: B. Threat ✓

Explanation: An attacker or entity with the potential to cause harm represents a threat.

 **Keyword:** Attacker / danger = Threat.

Q13. Risk

A company has an outdated server that can be exploited by attackers, potentially causing financial loss. The possibility of this loss occurring represents:

- A. Risk
- B. Encryption
- C. Authentication
- D. Vulnerability only

Answer: A. Risk ✓

Explanation: Risk is the potential for loss or impact when a threat exploits an existing vulnerability.

Q14. Vulnerability vs Threat

A server has an unpatched operating system vulnerability, and an attacker is attempting to exploit it. Which pairing is correct?

- A. Server flaw = Threat; attacker = Vulnerability
- B. Server flaw = Vulnerability; attacker = Threat
- C. Server flaw = Risk; attacker = Patch
- D. Server flaw = Threat; attacker = Risk

Answer: B. Server flaw = Vulnerability; attacker = Threat ✓

Explanation: The software flaw is the system weakness (vulnerability); the attacker is the potential source of harm (threat).

Q15. Patch Management

A vendor releases a security update that fixes a critical vulnerability in its software. The company's IT team tests and installs the update. This activity is part of:

- A. Patch management
- B. Phishing
- C. Spoofing
- D. Data mining

Answer: A. Patch management ✓

Explanation: Testing and applying software security updates to eliminate vulnerabilities is patch management.

 **Keyword:** Security update → Patch.

Q16. Least Privilege

A database administrator gives a reporting application permission to only read the required tables. The application cannot modify or delete data. Which principle is being followed?

- A. Zero Trust only
- B. Least privilege
- C. Open access
- D. Data redundancy

Answer: B. Least privilege ✓

Explanation: Granting only the bare minimum permissions required for a role or application embodies least privilege.

 **Keyword:** Minimum required permissions = Least Privilege.

Q17. Least Privilege Scenario

Why is the principle of least privilege important?

- A. It gives every user maximum access
- B. It reduces potential damage if an account or application is compromised
- C. It eliminates the need for authentication
- D. It makes all systems publicly accessible

Answer: B. It reduces potential damage if an account or application is compromised ✓

Explanation: By limiting permissions, attackers cannot access unneeded databases or execute administrative actions if an account is breached.

Q18. Defense in Depth

A company uses: Firewall + IDS/IPS + MFA + Antivirus + Encryption. Why is this approach called defense in depth?

- A. It uses multiple layers of security
- B. It uses only one strong firewall
- C. It removes authentication
- D. It eliminates network monitoring

Answer: A. It uses multiple layers of security ✓

Explanation: Defense in depth relies on multiple defensive layers so that if one control fails, others continue to protect assets.

 **Keyword:** Multiple security layers = Defense in Depth.

Q19. Security Logging

A server records: 09:30 — Successful login, 09:32 — Failed login, 09:35 — File accessed, 09:40 — Configuration changed. These records are called:

- A. Logs
- B. Encryption keys
- C. ACLs
- D. Certificates

Answer: A. Logs ✓

Explanation: Time-stamped records documenting system events and user actions are called logs.

Q20. Security Auditing

A security team reviews system logs and configurations to determine whether employees followed the company's security policies. What activity is this?

- A. Security auditing
- B. Encryption
- C. Packet routing
- D. Compression

Answer: A. Security auditing ✓

Explanation: Auditing is the systematic examination of configurations and activities against policy requirements.

Q21. Data Breach

An attacker gains unauthorized access to a company's customer database and steals names, addresses, and passwords. What has occurred?

- A. Data breach
- B. Patch management
- C. Authentication
- D. Honeypot

Answer: A. Data breach ✓

Explanation: Unauthorized access and exfiltration of protected/confidential information constitutes a data breach.

Q22. Data Breach Scenario

Which of the following could potentially lead to a data breach?

- A. Exploiting an unpatched vulnerability
- B. Phishing an employee
- C. Malware infection
- D. All of the above

Answer: D. All of the above ✓

Explanation: Breaches can occur due to vulnerabilities, social engineering (phishing), malware, misconfiguration, or insider threats.

Q23. 🔥 Mixed Scenario

A company follows these rules: Every user must verify their identity; Access is granted only when required; Users receive minimum necessary permissions; Suspicious activity is continuously monitored. Which combination of principles is most closely represented?

- A. Zero Trust + Least Privilege + Security Monitoring
- B. Open Access + Anonymous Access
- C. DAC + Email Bombing
- D. DDoS + Encryption

Answer: A. Zero Trust + Least Privilege + Security Monitoring ✓

Explanation: Verify identity & contextual access = **Zero Trust**; Minimum necessary permissions = **Least Privilege**; Continuous observation = **Security Monitoring**.

Q24. 🔥 Hard

A company discovers that its web server has an old software version containing a known security flaw. No attack has occurred yet, but an attacker could exploit the flaw. Which statement is most accurate?

- A. The software flaw is a vulnerability, while the potential attacker is a threat
- B. The software flaw is a threat, while the attacker is a vulnerability
- C. The software flaw is a data breach
- D. The attacker is a patch

Answer: A. The software flaw is a vulnerability, while the potential attacker is a threat ✓

Explanation: System flaw = **Vulnerability** (weakness); Potential attacker = **Threat** (danger source); Potential damage = **Risk**.

Q25. 🔥 Hard — Mixed Security Architecture

An organization wants: A decoy server to attract attackers; Centralized analysis of security logs; Minimum permissions for users; Multiple security layers; Regular security updates. Which combination is correct?

- A. Honeypot + SIEM + Least Privilege + Defense in Depth + Patch Management
- B. VPN + DNS + DHCP + NAT + FTP
- C. AES + RSA + DES + RC4 + RC5
- D. DoS + DDoS + Phishing + Spoofing + Ransomware

Answer: A. Honeypot + SIEM + Least Privilege + Defense in Depth + Patch Management ✓

Explanation: Decoy server → **Honeypot**; Centralized logs → **SIEM**; Minimum permissions → **Least Privilege**; Multiple layers → **Defense in Depth**; Regular updates → **Patch Management**.